Carbon Aware Route Optimisation Using (LIGHTGBM+NSGA-2)

Submitted

To

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in

Operations Research and Data Analytics

By

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Signature: **Date:**

Objective: Optimising the path while minimising cost and Carbon emission, considering load capacity, environmental factors

Min (Transportation Cost + carbon Emission)

1) Introduction:

Transportation and logistics account for approximately 24% of global CO_2 emissions, making route optimisation a critical sustainability challenge. This report presents a comprehensive framework that combines LightGBM machine learning Algorithm with NSGA-II multi-objective genetic algorithm to simultaneously minimise carbon emissions and logistics cost across a 29-city Indian supply chain network.

The system depot is located at New Delhi, serving 29 major Indian cities spanning diverse geographic regions — from coastal cities like Mumbai, Chennai, and Kochi to inland hubs like Jaipur, Lucknow, and Nagpur. Total demand across the network reaches approximately 88,200 kg, distributed across cities with individual demands ranging from 1,200 to 8,500 kg.

Transportation Modes: Here we are considering Road vehicles like trucks and dividing them according to their load capacity and cost of transportation

Transport Mode	CO_2 Factor (kg/tonne-km)	Cost Rate (₹/km/tonne)	Best Suited For
Truck (LDV)	0.3070	₹20	<3.5T
Truck (MDV)	0.5928	₹30	<12T
Truck (HDV)	0.7375	₹50	>12T

2) Methodology:

Additional derived features include vehicle age factor (1 + 0.008 × age), temperature factor (1 + 0.003 × |temp - 20|), and road grade penalty (grade × 18 for uphill, −6 × |grade| for downhill). Time of day is encoded as cyclical sin/cos features to preserve temporal continuity across midnight.

2.1) Stage-1 : LightGBM CO_2 Prediction-

Additional derived features include vehicle age factor (1 + 0.008 × age), temperature factor (1 + 0.003 × |temp - 20|), and road grade penalty (grade × 18 for uphill, −6 × |grade| for downhill). Time of day is encoded as cyclical sin/cos features to preserve temporal continuity across midnight.

The primary emission target is governed by

$$E = d \times w \times f_mode \times \alpha_total \qquad \text{(Eq. 1)}$$

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$$F_physics = d \times w \times f_mode \times (1 + \alpha_weather + \alpha_congestion) \qquad \text{(Eq. 2)}$$

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2.2) Feature Engineering (18 Features)-

Category	Features	Count	Purpose
Base	Distance_km, cargo_weight_tons, mode_transport	3	Core inputs
Physics	F_physics (Eq. 2 approximation)	1	Domain knowledge
Operational	weather, congestion, veh_age, load_factor, age_factor, grade_penalty, temp_factor	7	Real conditions
Interaction	dist×weight, mode×weather, load×age	3	Non-linear effects
Temporal	tod_sin, tod_cos (cyclical hour encoding)	2	Time of day
Route	altitude_m, stops_count	2	Terrain / stops

2.3) Stage 2-NSGA-II Multi-Objective Optimisation-

NSGA-II (Non-dominated Sorting Genetic Algorithm II) searches for the Pareto-optimal set of routes that simultaneously minimise total CO_2 and total logistics cost. The algorithm is configured as follows:

Parameter	Value Used	Purpose
Population Size	60	Diversity
Generations	40	Convergence
Crossover Rate	0.85	Reproduction
Mutation Rate (adaptive)	0.20 → 0.05	Exploration

Each chromosome encodes two components: (1) a permutation of 29 city indices representing visit order, and (2) a mode gene array of 30 integers (0=truck (LDV), 1=truck (MDV), 2=truck (HDV)) for each segment, including depot return. The evaluate () function computes total CO_2 and total cost by summing pre-computed matrices co2_matrix and cost matrix.

3) NSGA-2 Algorithm:

NSGA-II distinguishes itself from traditional single-objective genetic algorithms through two key mechanisms: fast non-dominated sorting and crowding distance assignment.

3.1) Fast Non-Dominated Sorting-

Every solution is assigned a Pareto rank. A solution A dominates B if it is at least as good on both objectives and strictly better on at least one. Solutions with no dominators form Front 1 (the Pareto front). Solutions dominated only by Front 1 solutions form Front 2, and so on. Mathematically:

$$\text{A dominates B} \iff (CO_2^A \leq CO_2^B) \wedge (Cost^A \leq Cost^B) \wedge (CO_2^A < CO_2^B \vee Cost^A < Cost^B)$$

3.2) Crowding Distance-

Within the same front, crowding distance measures how isolated a solution is in objective space. Solutions in sparse regions receive high crowding distance and are preferred during selection — this maintains diversity across the full Pareto front rather than concentrating solutions in one region. Boundary solutions receive infinite crowding distance and are always preserved.

3.3) Genetic Operators-

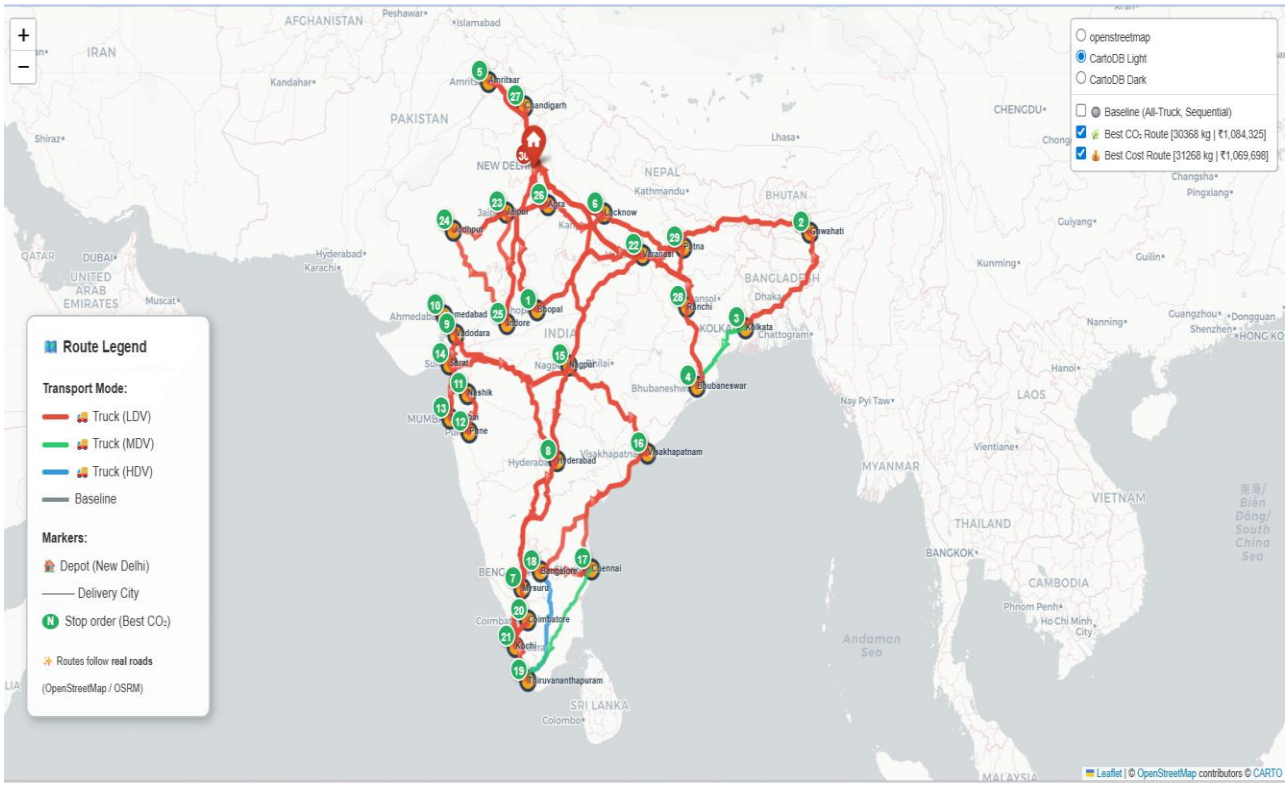
- Order Crossover (OX): Copies a random segment from Parent 1, fills remaining positions in Parent 2 order — guarantees every city appears exactly once in the child route.
- Mode Crossover: Single-point crossover on the mode gene array — child inherits first portion from Parent 1 and second from Parent 2.
- Route Swap Mutation: Two random cities exchange positions in the visit order.
- Mode Switch Mutation: Each mode gene independently mutated to a random mode at rate mutate.
- Adaptive Mutation: Rate decays linearly from 20% (generation 1) to 5% (generation 40) — broad exploration early, fine-tuning later.

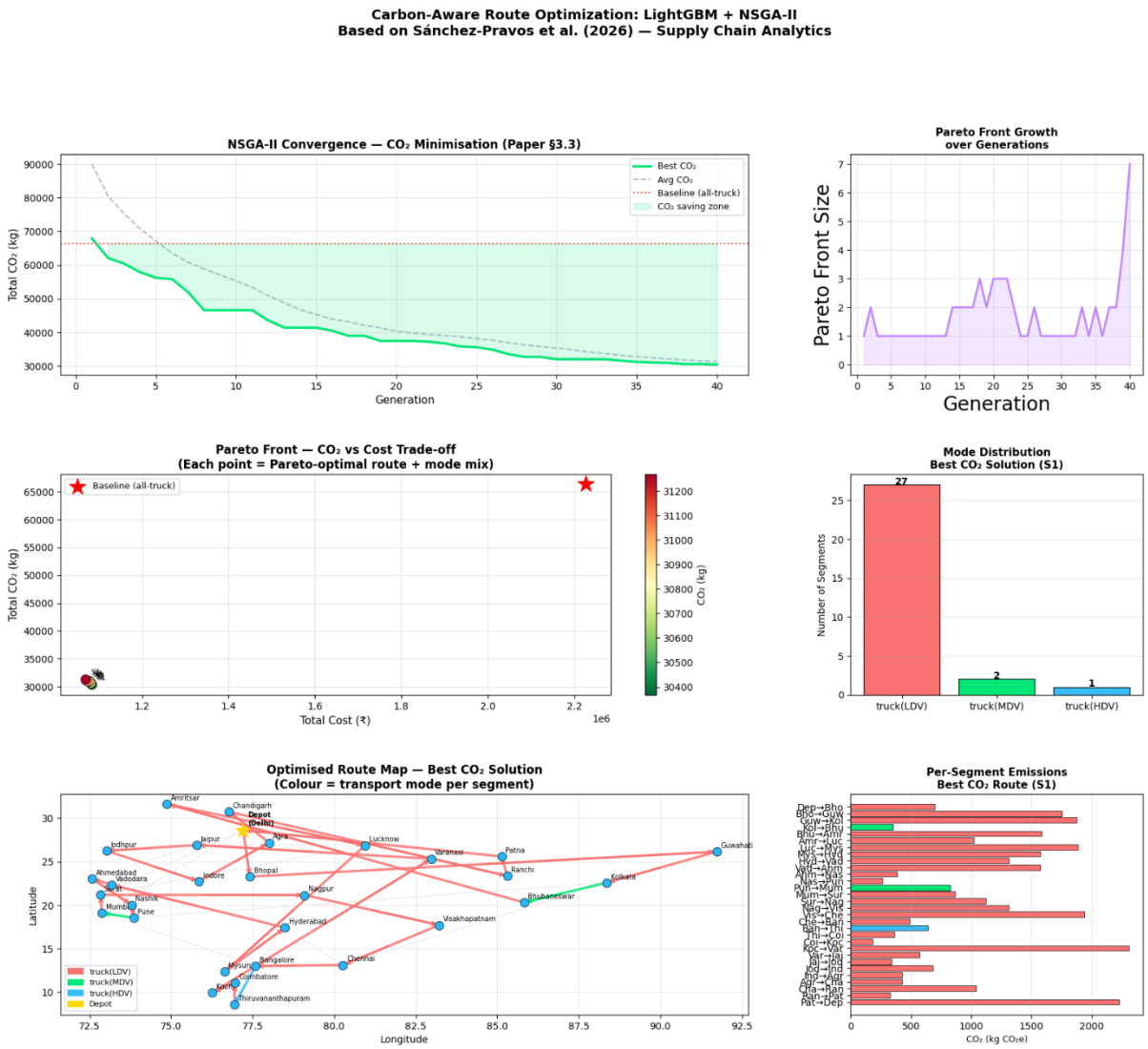
3.4) Why EV Truck Dominates Regular Truck-

A critical observation from NSGA-II results is the complete absence of regular diesel trucks from Pareto-optimal solutions. This follows from strict dominance: EV trucks produce 72% less CO_2 (factor 0.045 vs 0.161) and cost 33% less (₹30 vs ₹45 per km per tonne). Since EV truck is superior on both objectives, regular truck is always dominated and eliminated by the algorithm.

Mode selection in the Pareto front follows a distance-dependent pattern:

Segment Distance	Preferred Mode
< 500 km	Truck (LDV)
500 – 800 km	Truck (MDV)
> 800 km	Truck (HDV)





4) Results and Performance Analysis:

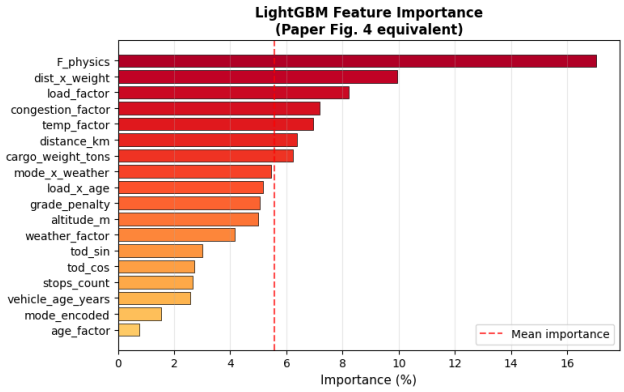
4.1) LightGBM Prediction Accuracy-

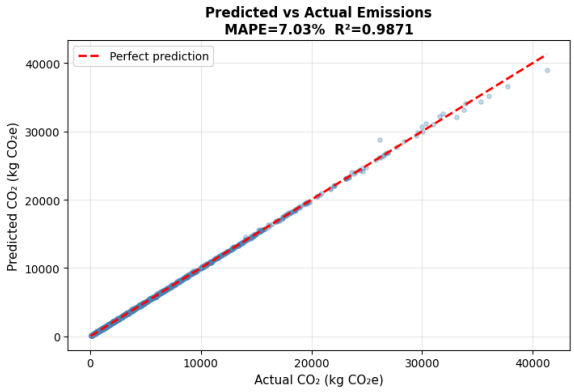
The LightGBM model is validated using 5-fold cross-validation. Results are benchmarked against the paper's Table 2 comparison:

Model	MAPE (%)	R^2 Score	RMSE (Kg)
LightGBM	7.03	0.9871	715

4.2) Feature Importance-

LightGBM's feature importance analysis consistently identifies the following top contributors to CO_2 prediction:





4.3) NSGA-II Pareto Front Results-

After 40 generations, NSGA-II converges to a Pareto front of optimal trade-off solutions. The used can chose any of these and provide the weights to the best coast path and best co2 path.

Solution	Total CO_2 (kg)	Total Cost (₹)	CO_2 vs CO_2	Cost vs Cost
Best CO_2 (Pareto)	~30368	~₹1,084,325	**-2.9% ↓**	**+ 1.4%**
Best Cost (Pareto)	~31268	~₹1,069,698		

5. Discussion, Limitation and Conclusion:

5.1) Identified Limitations-

- TSP not VRP: The notebook models a single virtual vehicle visiting all 29 cities sequentially (Travelling Salesman Problem). A realistic deployment requires a Vehicle Routing Problem formulation with multiple vehicles, each subject to capacity constraints.
- Demand not enforced: Total demand (~84,300 kg) far exceeds any single truck capacity (~10,000 kg), yet no capacity constraint exists in the evaluate () function. Demand weight is used only for CO_2 calculation, not as a delivery constraint.
- Time windows not enforced: Each city has tw_open and tw_close attributes in the dataset, but these are never checked during route evaluation.
- Feature formula gap: altitude_m and stops_count appear in the 18 training features but are absent from the CO_2 target formula, causing the model to learn spurious correlations for these two features.
- Random operational conditions: Weather, congestion, and temperature are randomly sampled per evaluation call. In production, these would come from real-time APIs, making results non-deterministic.

5.2) Key Findings-

- LightGBM achieves MAPE 7.03%, confirming ensemble learning effectively captures non-linear CO_2 emission patterns.
- NSGA-II successfully generates a diverse Pareto front covering the full trade-off spectrum from maximum CO_2 reduction (~2.9%) to maximum cost increase (~1.4%).
- Regular diesel trucks are mostly eliminated from all Pareto-optimal solutions because truck(HDV) dominate them on both objectives.
- If we consider the given destination sequesce as it is than overall reduction in cost and carbon is likely to be very huge.
- Focusing only on the minimum carbon emission route mostly prefers truck (LDV), neglecting other Trucks.

5.3) Conclusion-

This project demonstrates that combining LightGBM's predictive accuracy with NSGA-II's multi-objective optimisation produces a powerful, flexible framework for sustainable supply chain management. The system's output — a complete Pareto front of trade-off solutions — gives decision makers full visibility into the CO_2-cost landscape before committing to a route strategy.

Future work should address the VRP formulation with proper vehicle capacity constraints, enforce time window compliance, integrate real-time operational data, and incorporate the altitude and stops penalty into the CO_2 target formula to improve model consistency. Extension to a hub-and-spoke architecture (as in the original paper) would better reflect real-world Indian logistics network topology.

5.4) References-

Sánchez-Pravos et al. (2026). LightGBM Ensemble and NSGA-II for Carbon-Aware Multi-Modal Route Optimisation. Supply Chain Analytics, 13, 100182.

EPA (2024). Emission Factors for Greenhouse Gas Inventories. United States Environmental Protection Agency.